

Project Topics

1. Differential Privacy and Online Learning

Differential privacy limits how much an algorithm’s output can reveal about any one person’s data. It requires similar output distributions on datasets that differ in a single record. This is a form of stability, which also appears in regret analysis when we bound how much a learner’s decisions change after a new loss. The connection helps explain why techniques such as adding random perturbations are useful in both private learning and no-regret algorithms. Privacy can also be a constraint on the online problem itself, raising questions about the tradeoff between privacy and regret.

- Jacob Abernethy, Young Hun Jung, Chansoo Lee, Audra McMillan, and Ambuj Tewari. *Online Learning via the Differential Privacy Lens* (NeurIPS 2019).
- Alon Gonen, Elad Hazan, and Shay Moran. *Private Learning Implies Online Learning: An Efficient Reduction* (NeurIPS 2019).

2. Fair Classification via No-Regret Learning

Fair classification studies how to minimize prediction error while imposing constraints such as similar false-positive rates across demographic groups. The problem becomes harder when these constraints must hold for many overlapping groups. One approach is to formulate a game: a learner chooses a classifier, while an auditor searches for violated constraints. No-regret dynamics then give a way to find classifiers with guarantees on both error and fairness. This is an application of minimax and constrained optimization, with additional questions about how to audit fairness and how to solve the optimization problems faced by each player.

- Alekh Agarwal, Alina Beygelzimer, Miroslav Dudík, John Langford, and Hanna Wallach. *A Reductions Approach to Fair Classification* (ICML 2018).
- Michael Kearns, Seth Neel, Aaron Roth, and Zhiwei Steven Wu. *Preventing Fairness Gerrymandering: Auditing and Learning for Subgroup Fairness* (ICML 2018).

3. Human–AI Collaboration and Agreement

Human experts and prediction models may have different information about the same problem. Exchanging predictions can help them make use of what the other knows, but an agreement is only useful if the resulting predictions are accurate. Calibration offers a way to study this exchange: predicted probabilities should match observed frequencies, including on cases selected using the other party’s predictions. Disagreement can then expose systematic errors that call for a correction. In repeated prediction, no-regret learning provides tools for designing these updates and comparing the joint predictions with what either participant could achieve alone.

- Natalie Collina, Surbhi Goel, Varun Gupta, and Aaron Roth. *Tractable Agreement Protocols* (STOC 2025).
- Natalie Collina et al. *Collaborative Prediction: Tractable Information Aggregation via Agreement* (SODA 2026).

4. AI Alignment: Preference Optimization and Welfare

Training an AI system from human preferences requires deciding how to combine judgments from people who may disagree. Pairwise comparisons are useful feedback, but winning more comparisons need not mean that a model serves users well on average. One approach treats policies for generating responses as players in a zero-sum game and uses an equilibrium to choose a policy. This brings in no-regret learning and mirror descent. It also connects to social choice theory: we can ask how much average user utility is lost when an objective is based on comparisons that give only limited information about users' underlying utilities.

- Paul Gözl, Nika Haghtalab, and Kunhe Yang. *Distortion of AI Alignment: Does Preference Optimization Optimize for Preferences?* (NeurIPS 2025).
- Rémi Munos et al. *Nash Learning from Human Feedback* (ICML 2024).

5. Multi-Distribution Learning

Multi-distribution learning seeks a model that performs well across several populations or data sources. For example, a model used by several hospitals might be evaluated by its worst expected loss at any hospital. This is a minimax objective: the learner chooses a predictor, and an adversary chooses a distribution on which to evaluate it. When the learner can request data from each source, it must also decide where to spend its sampling budget. No-regret dynamics and adaptive sampling help solve this game while controlling the total number of examples needed.

- Nika Haghtalab, Michael I. Jordan, and Eric Zhao. *On-Demand Sampling: Learning Optimally from Multiple Distributions* (NeurIPS 2022).
- Zihan Zhang, Wenhao Zhan, Yuxin Chen, Simon S. Du, and Jason D. Lee. *Optimal Multi-Distribution Learning* (COLT 2024).

6. Incentive Design for Learning Agents

Incentive design asks how to influence agents who are learning from experience. A designer may offer payments or contracts to encourage a particular equilibrium or increase its own payoff. The agents' responses depend on their learning algorithms and their history of payoffs, so the designer must reason about behavior over time. Regret guarantees provide a way to analyze these responses without specifying every detail of an agent's algorithm. Questions include how much the designer must pay, what it needs to observe, and whether the average payment can vanish as the interaction continues.

- Brian Hu Zhang et al. *Steering No-Regret Learners to a Desired Equilibrium* (EC 2024).
- Guru Guruganesh et al. *Contracting with a Learning Agent* (NeurIPS 2024).

7. Online Conformal Prediction

Online conformal prediction studies how to produce prediction intervals with reliable coverage as data change over time. For example, we may want the observed outcome to lie in the predicted interval on 90% of rounds, while keeping the intervals reasonably narrow. Coverage errors give feedback for adjusting future intervals. This turns uncertainty quantification into an online learning problem, with the step size controlling how quickly the method reacts to changes. The connections to online gradient descent and adaptive regret help us understand both long-run coverage and performance over shorter periods.

- Isaac Gibbs and Emmanuel Candès. *Adaptive Conformal Inference Under Distribution Shift* (NeurIPS 2021).
- Isaac Gibbs and Emmanuel J. Candès. *Conformal Inference for Online Prediction with Arbitrary Distribution Shifts* (Journal of Machine Learning Research, 2024).

8. Counterfactual Regret Minimization (CFR)

Counterfactual regret minimization is a way to learn strategies in sequential games with hidden information, such as poker. A player's decision depends on an information set: the game histories that are indistinguishable to that player. CFR handles the large strategy space by running a local regret minimizer at each information set. A regret decomposition connects these local guarantees to regret for the full game. In two-player zero-sum games with perfect recall, where players remember their own past observations and actions, this gives average strategies that approach Nash equilibrium. Sampling methods can reduce the work needed to update the local learners.

- Martin Zinkevich, Michael Johanson, Michael Bowling, and Carmelo Piccione. *Regret Minimization in Games with Incomplete Information* (NIPS 2007).
- Marc Lanctot, Kevin Waugh, Martin Zinkevich, and Michael Bowling. *Monte Carlo Sampling for Regret Minimization in Extensive Games* (NIPS 2009).

9. Omniprediction

Omniprediction asks for a single predictor that can be used with many different loss functions. Users who assign different costs to false positives and false negatives, for example, may choose different actions from the same predicted probability. The aim is to support these choices without retraining for each loss, while competing with the best predictor in a specified class. Multicalibration is one route to such guarantees: it checks for systematic errors detectable by that class among examples receiving similar predictions. In an online setting, the objective becomes simultaneous low regret under many loss functions, connecting omniprediction to calibration and regret minimization.

- Parikshit Gopalan, Adam Tauman Kalai, Omer Reingold, Vatsal Sharan, and Udi Wieder. *Omnipredictors* (ITCS 2022).
- Sumegha Garg, Christopher Jung, Omer Reingold, and Aaron Roth. *Oracle Efficient Online Multicalibration and Omniprediction* (SODA 2024).